How is Helmholtz electrical double layer formed?

When a metal is in contact with its own salt solution either oxidation or reduction occurs depending on the nature of the metal. Thus electrode gets negative or positive charge and the solution gets the opposite charge. So at equilibrium an electrical double layer is formed. This electrical double layer is known as Helmholtz electrical double layer.

Calculate the single electrode potential of dichromate electrode at 25°C when $[Cr_2O_7^{2-}]$ is 0.3M, $[Cr^{3+}]$ is 0.02M and $[H^+]$ is 1M. ~~Given~~ $\cancel{Cr_2O_7^{2-} + 14H^+ + 6e \rightarrow}$

Given $Cr_2O_7^{2-} + 14H^+ + 6e \rightarrow 2Cr^{3+} + 7H_2O$; $E^\circ = 1.33V$

~~Given~~ Given cell reaction

$$Cr_2O_7^{2-} + 14H^+ + 6\bar{e} \longrightarrow 2Cr^{3+} + 7H_2O$$

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{[Cr^{3+}]^2}{[Cr_2O_7^{2-}][H^+]^{14}}$$

$$E_{cell} = 1.33 - \frac{0.0591}{6} \log \frac{[0.02]^2}{[0.3][1]^{14}}$$

$$E_{cell} = 1.358V$$

3. Recognize the atoms showing NMR phenomenon among the following. Give reason. a) $^{1}_{1}H$ b) $^{2}_{1}H$ c) $^{3}_{1}H$ d) $^{16}_{8}O$ e) $^{14}_{7}N$ f) $^{18}_{8}O$

→ a) In $^{1}_{1}H$

$N_p = 1$, $N_n = 0$

i.e, N_p is odd and N_n is even

Due to half integral spin, $^{1}_{1}H$ is NMR active.

b) $^{2}_{1}H$

$N_p = 1$ $N_n = 1$

i.e, N_p and N_n are odd

Due to integral spin, $^{2}_{1}H$ is NMR active

c) $^{3}_{1}H$

$N_p = 1$ $N_n = 3 - 1 = 2$

i.e, N_p is odd, N_n is even

Due to integral spin, $^{3}_{1}H$ is NMR active

d) $^{16}_{8}O$

$N_p = 8$ $\quad N_n = 16 - 8 = 8$

i.e, N_p and N_n are even

$^{16}_{8}O$ is NMR inactive.

e) In $^{14}_{7}N$

$N_p = 7$ $\quad N_n = 14 - 7 = 7$

i.e, N_p and N_n are odd

Due to integral spin, $^{14}_{7}N$ is NMR active.

f) $^{18}_{8}O$

$N_p = 8$ $\quad N_n = 18 - 8 = 10$

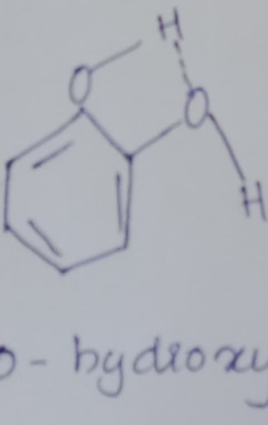
i.e, N_p and N_n are even

$^{18}_{8}O$ is NMR inactive

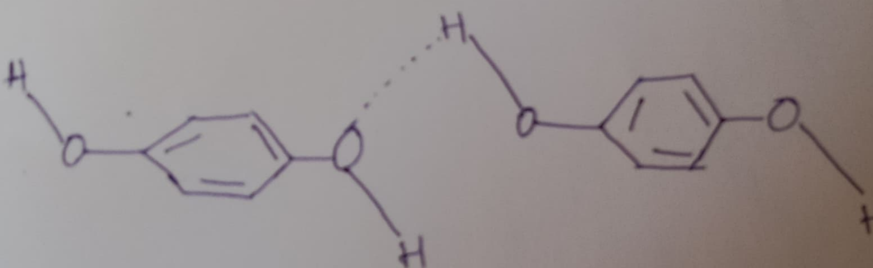
4. IR spectroscopy can be used to differentiate intra molecular and inter molecular hydrogen bonds. Explain with an example

IR spectroscopy can be used to differentiate intra molecular and intermolecular hydrogen bonds. The – OH stretching frequency (3300 – 3500 cm) does not change on dilution in the case of intra-molecular H- bonding (Eg: O-hydroxy phenol).

In P-hydroxy phenol, there exist intermolecular hydrogen bonding. On dilution, the molecules get separated and intermolecular H- bonding breaks and there occur a shift in absorption frequency with dilution.



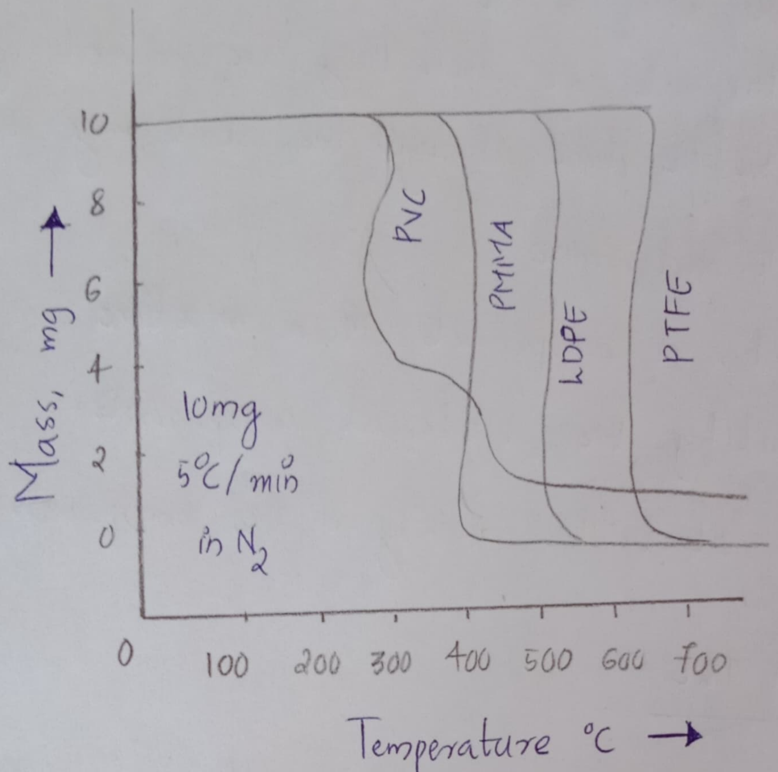
o- hydroxy phenol (intra molecular H-bonding)



P-hydroxy phenol - inter molecular H-bonding

Solved Question - December 2021 Paper

5). Compare the thermal stability of PVC, PMMA, LDPE and PTFE using TG given below. Justify your answer.



The curve clearly indicates that PVC is the least thermally stable and PTFE is the most thermally stable. The graph shows that PVC starts decomposition at low temperature compared to LDPE (Low Density Poly Ethylene). This is due to the elimination of HCl in PVC. PTFE possess high thermal stability due to strong C-F bond than C-H bond in other polymer. i.e, the order of thermal stability is PVC < PMMA < LDPE < PTFE.

6). Give the principle of TLC. Mention two applications of TLC.

Principle :- The components of a mixture are separated depending on the extent of adsorption and the rate at which they are carried by the moblie phase. through the stationary phase.

- It is used to determine
  * Number of components in a mixture
  * Identify the different components.
  * To detect the purity of the compound.

Application of TLC

(a). Used to check the purity of sample
(b). Used to check adulteration in food.
(c). Used to determine appropriate solvent for column chromatography.

A sample of water on analysis gives the following results. Ca^{2+} = 200 mg/L, Mg^{2+} = 180 mg/L, HCO_3^- = 360 mg/L, Cl^- = 200 mg/L and Na^+ = 80 mg/L. Calulate temporary and permanent hardness.

Total hardness is due to Ca^{2+} and Mg^{2+} ions.

Temporary hardness is due to bicarbonates ions.

Permanent hardness = Total hardness - Temporary hardness.

$$\text{Total hardness} = 200 \times \frac{100}{40} + 180 \times \frac{100}{24}$$

$$= 500 + 750$$

$$= 1250 \text{ ppm}$$

$$\text{Temporary hardness} = 360 \times \frac{100}{122}$$

$$= 295.081 \text{ ppm}$$

$$\therefore \text{permanent hardness} = 1250 - 295.081$$

$$= 954.919 \text{ ppm}$$

Differentiate between aerobic and anaerobic oxidation.

Aerobic oxidation	Anaerobic oxidation
• Takes place in presence of excess O_2.	• Takes place in presence of insufficient quantity of O_2.
• oxidation by aerobic bacteria.	• oxidation by anaerobic bacteria.
• products of oxidation are CO_2, nitrates, phosphates, sulphates.	• products of oxidation are acetic acid, CH_4, H_2S, phosphene, NH_3.
• During decomposition carbohydrates are converted to CO_2 and H_2O.	• During decomposition carbohydrates are converted to CO_2 and CH_4
• No biogas fuel is produced.	• Biogas fuel like CH_4 is produced.
• Energy released more and rate of decomposition slow.	• Energy released less and rate of decomposition fast.